



Fox School of Business

TEMPLE UNIVERSITY®

Department of Risk, Actuarial, Healthcare Management, Legal Studies
AS 3504/5108 Actuarial Analytics

Practice Exam #2 (AS 3504) Solutions

Student Name: _____

Score: _____ / _____

75 Minutes. 40 Points.

Show all your work on this exam.

The more I understand your thought process, the more partial credit I can give, but writing a bunch of formulas down on a Q you have no idea how to start is not work.

No work = No credit for numerical questions.



Conceptual Questions Section (1 Point Each x 15). Clearly circle the True statements.

1. Multiple linear regression assumes the response terms are independent and normally distributed with mean 0 and constant variance σ^2 .
2. In multiple linear regression, adding any new predictor will never decrease R^2 .
3. As the number of data points in a regression model approaches infinity, the width of the prediction interval approaches 0.
4. Adding an interaction term between a categorical variable with 5 levels and another categorical variable with 4 levels requires 12 additional parameters.
5. In a t -test of a parameter in MLR, the higher the t -statistic, the stronger the evidence is to reject the null hypothesis and decide to reject the parameter.
6. Forward stepwise selection cannot be used when the data has more parameters than observations.
7. k -fold cross-validation leads to greater variance than leave-one-out-cross-validation for $k < n$.
8. Mallows's C_p and AIC are identical metrics that produce the same model selection outcomes.
9. A residual plot can help identify outliers and high leverage points.
10. Possible approaches to addressing heteroscedasticity include transforming the response or making no adjustments.
11. Heteroscedasticity causes the estimate of mean squared error (MSE) to be unreliable.
12. Orthogonal predictors guarantee the absence of multicollinearity.
13. Cook's distance combines information about both residual size and leverage to identify influential observations.
14. Multicollinearity can make individual coefficient t -tests harder to reject, even when the model's predictive performance is not necessarily harmed.
15. A partial F -test can compare any two models to test whether a group of terms adds significant explanatory value.



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Conceptual Questions Solutions:

1. False: error terms
2. True
3. False; always has variability from prediction error
4. True
5. False; keep the parameter
6. False; backwards cannot
7. False; lower
8. False; proportional metrics that do produce the same selection
9. False; outliers but not high leverage
10. True
11. True
12. True
13. True
14. True
15. False; nested only



Computation Questions Section (5 Points Each x 5)

16. The # of wins of 32 hockey teams are studied using a multiple linear regression. All teams belong to these divisions: I, II, or III. You are given the following results:

	Coefficients
Intercept	8
Coach Experience	0.2
I	-1.6
II	-0.6

- A. (3 points) What is the coach experience for a Division II team predicted to have 8.6 wins?
- B. (2 points) Is it possible for a team in Division I to have greater predicted # of wins than a team from Division III? Explain why or why not and use some numerical evidence in your justification.

Sol: MLR

$$8.6 = 8 - 0.6 + 0.2x$$

$$x = 6$$

Division I can have more predicted wins than III if there is an experience gap of $\frac{1.6}{0.2} = 8$ or more in favor of the Division I team's Coach Experience.



17. The following models are fitted to 30 observations:

I: $Y = \beta_0 + \beta_1 X_1 + \varepsilon$

II: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$

$$\sum (Y_i - \bar{Y})^2 = 160$$

$$\sum (X_1 - \bar{X}_1)^2 = 10$$

In Model I, $b_1 = -2$.

In Model II, $R^2 = 0.70$.

A. (4 points) Determine the value of the F -statistic used to test that β_2 and β_3 are jointly equal to zero.

B. (1 point) What does a full-model F -test help determine?

Sol:

MLR Inference

$$SSReg_I = (-2)^2 * 10 = 40$$

$$SSE_I = 160 - 40 = 120$$

$$SSReg_{II} = 0.7(160) = 112$$

$$SSE_{II} = 160 - 112 = 48$$

$$F = \frac{\frac{120 - 48}{2}}{\frac{48}{30 - (3 + 1)}} = 19.5$$

Whole model F -test assesses if a model overall is statistically significant in predicting the variance of the dependent variable.



18. A linear regression model predicts the DPS of a game character. The model's residual standard error was 50.

Parameter	Coefficient Estimate	Standard Error
Intercept	100	20
Level	10	4
WizardYes	200	125
HP	-3	0.5

- A. (4 points) Find the Cook's distance of a Level 9, 50 HP, Wizard character with leverage of 0.07 and DPS of 200.
- B. (1 point) What is the typical standard for a highly influential point?

Sol:

Model Diagnostics

$$\hat{y}_i = 100 + 10 * 9 + 200 - 3 * 50 = 240$$

$$e_i = 200 - 240 = -40$$

$$D = \frac{(-40)^2(0.07)}{50^2(3 + 1)(1 - 0.07)^2} = 0.013$$

3x the average Cook's Distance which is approximately $\frac{1}{n}$



19. For a multiple linear regression model, the following statistics are provided

$$X = \begin{bmatrix} 1 & 0 & 1 & 9 \\ 1 & 1 & 1 & 15 \\ 1 & 1 & 1 & 8 \\ 0 & 1 & 1 & 7 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & 6 \end{bmatrix}; \quad y = \begin{bmatrix} 21 \\ 32 \\ 19 \\ 17 \\ 15 \\ 15 \end{bmatrix}$$

$$X(X^T X)^{-1} X^T = \begin{bmatrix} 0.684 & 0.070 & 0.247 & -0.171 & -0.146 & 0.316 \\ 0.070 & 0.975 & -0.044 & 0.108 & -0.038 & -0.070 \\ 0.247 & -0.044 & 0.797 & 0.063 & 0.184 & -0.247 \\ -0.171 & 0.108 & 0.063 & 0.418 & 0.411 & 0.171 \\ -0.146 & -0.038 & 0.184 & 0.411 & 0.443 & 0.146 \\ 0.316 & -0.070 & -0.247 & 0.171 & 0.146 & 0.684 \end{bmatrix}$$

$$(X^T X)^{-1} X^T y = \begin{bmatrix} 0.297 \\ -0.032 \\ 3.943 \\ 1.854 \end{bmatrix}; \quad X(X^T X)^{-1} X^T y = \begin{bmatrix} 20.93 \\ 32.03 \\ 19.04 \\ 16.89 \\ 15.04 \\ 15.07 \end{bmatrix}; \quad \hat{\sigma}^2 = 0.012657$$

Any matrix provided on the exam will be clearly labeled as to its meaning.

The $X(X^T X)^{-1} X^T$ is the hat matrix.

The $X(X^T X)^{-1} X^T y$ is the prediction vector.

- (2 points) What is the largest leverage?
- (2 points) What is the largest magnitude standardized residual?
- (1 point) Are there any high leverage points or outliers in this dataset?

Sol:

Model Diagnostics

A. The $X(X^T X)^{-1} X^T$ is the hat matrix and the diagonal entries are the leverage by observation. The highest is 0.975 and the threshold is $\frac{3(p+1)}{n} = \frac{3(3+1)}{6} = 2$ so is not high leverage.

B. $X(X^T X)^{-1} X^T y$ represents the prediction vector, and the largest residual is the 4th observation of $17 - 16.89 = 0.11$.

$$e_{4,std} = \frac{e_i}{s\sqrt{1-h_{ii}}} = \frac{0.11}{\sqrt{0.012657} \cdot \sqrt{1-0.418}} = 1.28 < 2 \text{ so is not an outlier}$$

C. Because the largest leverage is not high leverage and largest standardized residual is not an outlier, there are none.



20. The following dataset with 75 observations is used to predict capacity at a hotel.

Weekend is a categorical variable with 2 levels.

Special is a categorical variable with 6 levels.

The rest are continuous variables.

A. (2 points) Calculate the p needed for each model.

B. (3 points) Which model is selected based on adjusted R^2 ? Show your calculations that justify your selection.

Model	Predictors	R^2
I	Income, Temp	60%
II	Income, Weekend	61%
III	Income, Special	63%
IV	Income, Temp, Weekend	62%
V	Income, Temp, Special	64%

Sol:

Model Selection

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \left(\frac{n-1}{n-p-1} \right)$$

Model	p	Adjusted R^2
I	2	Not A Winner
II	2	59.9%
III	6	59.7%
IV	3	60.4%
V	7	60.2%

Model IV is the best.